Electronic Circuits I

Practice midterm **90 minutes | 30 marks**

Scope: Lectures 1–8 and Sheets 1–7, including the additional op-amp problems, DACs and ADCs. This is an original practice paper, not an official or predicted exam.

Name: _____ **ID:** _____

Instructions. Answer all five questions on separate answer sheets. Show derivations and units. A scientific calculator is allowed. Ideal op-amps are assumed unless stated otherwise; assume adequate supply rails. Treat coupling and bypass capacitors as open circuits for DC and shorts for midband AC. Use $V_T = 25$ mV. State any additional approximation.

Question	Marks	Suggested time
1. Op-amp synthesis and limits	5	15 min
2. DAC and ADC	7	21 min
3. Voltage regulation	3	9 min
4. 555 timer	4	12 min
5. BJT bias and amplifiers	11	33 min

Question 1. Op-amp synthesis and limits [5 marks]

(a) Using **at most three ideal op-amps**, design a circuit implementing

$$v_o(t) = (1\,\text{ms})\frac{dv_1}{dt} + (100\,\text{s}^{-1})\int_0^t v_2(\tau)\,d\tau - 2v_3(t).$$

Assume zero initial capacitor voltage. Draw the complete circuit, label all components, and give one valid set of resistor and capacitor values. Justify the signs of all three terms. **[3]**

(b) A noninverting amplifier has $R_f = 90\,\text{k}\Omega$, $R_g = 10\,\text{k}\Omega$, and finite open-loop gain $A_0 = 1000$. Derive its closed-loop gain and calculate its percentage reduction relative to the ideal gain. **[1]**

(c) A conventional inverting BJT log amplifier uses an input resistor and a transistor feedback element that carries positive current for $v_{\text{in}} > 0$. Explain why its logarithmic formula cannot be extended to a negative input voltage. **[1]**

Use the remaining time after each question to check signs, operating regions, and loading.

Question 2. Digital/analog conversion [7 marks]

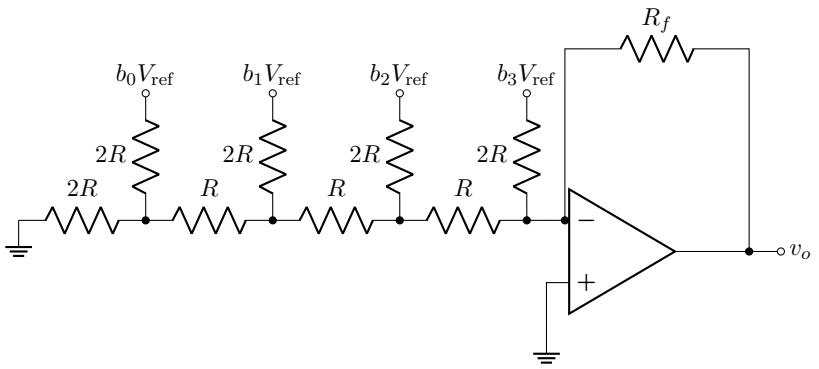


Figure 1. Four-bit R–2R DAC; b_3 is the MSB.

In Figure 1, each bit b_i is either 0 or 1; its input terminal is connected to b_iV_{ref}. Assume ideal switches.

(a) With $(b_3b_2b_1b_0) = 0010$, derive v_o using successive Thévenin reductions or equivalent current division. Then write the general expression for v_o for any input code. **[2]**

(b) Choose R, $2R$, R_f and a **positive** V_{ref} so the resistance seen at the MSB terminal is 20 kΩ and the output spans exactly 0 to −3.75 V. Use $R_f = R$. State the signed one-LSB step and the output at code 1011. **[2]**

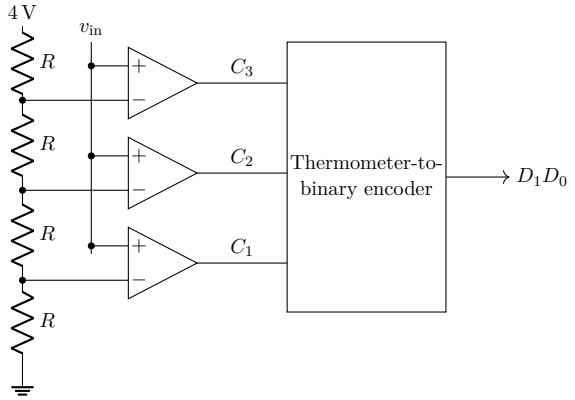


Figure 2. Two-bit flash ADC using ideal op-amps as comparators.

The divider uses four equal resistors. The encoder outputs the binary count of comparators that are high. Define $C_k = 1$ when $v_{\text{in}} \geq k$ V, for $k = 1, 2, 3$.

(c) Complete the table and sketch D_1 and D_0 versus sample number, holding each code until the next sample. **[2]**

Sample n	1	2	3	4	5	6
v_{in} (V)	0.4	1.2	2.6	3.4	2.2	0.8
$C_3C_2C_1$						
D_1D_0						

(d) State the ADC step size and the number of comparators required for a 3-bit version. Explain why $v_+ = v_-$ must **not** be imposed on these comparator op-amps. **[1]**

Question 3. Voltage regulation [3 marks]

A 7812 regulator is assisted by a PNP external pass transistor. Its emitter connects to $V_{\text{in}} = 18$ V, its collector to the regulated output, and its base to the IC input. A resistor R_{ext} connects the emitter to the base. The IC ground pin is grounded. Use $V_{EB} = 0.70$ V; neglect base and ground-pin currents. In the simplified current-sharing model, the IC carries 0.40 A once the pass transistor turns on.

(a) State the nominal output voltage and select R_{ext} so the external transistor starts conducting at an IC current of 0.40 A. **[1]**

(b) With $R_L = 12\,\Omega$, calculate the load current and external-transistor dissipation. **[1]**

(c) Explain why the IC's internal current limit does not by itself protect the external transistor. Describe a modification that limits the *total* load current to 1.5 A, and calculate the sense resistor for a 0.70 V turn-on threshold. **[1]**

Question 4. 555 timer design [4 marks]

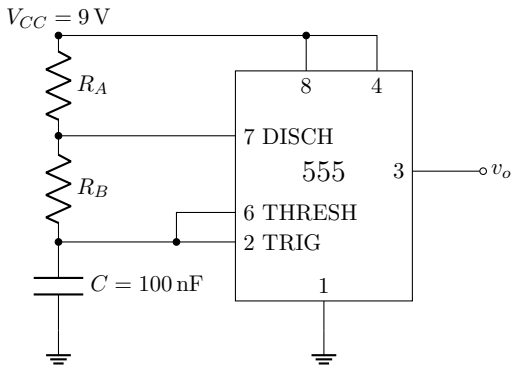


Figure 3. Standard astable connection; pin-5 decoupling is omitted.

(a) State the lower and upper capacitor switching voltages. Design R_A and R_B for $f = 1$ kHz and a high-state duty cycle of 75%, using $C = 100$ nF. Show the timing equations. **[2]**

(b) Redesign for the same frequency and capacitance but a duty cycle of 40%. Draw the necessary modification, specify diode orientation, and calculate the new resistor values. Use an ideal diode and ideal discharge switch. Explain why the unmodified circuit cannot achieve 40%. **[2]**

Question 5. BJT bias and small-signal analysis [11 marks]

Parts (a)–(d): Use Figure 4. Take $\beta = 100$, $V_{BE} = 0.70$ V for the DC calculation, and neglect the Early effect. The input v_i is supplied by an ideal AC voltage source through C_{in}.

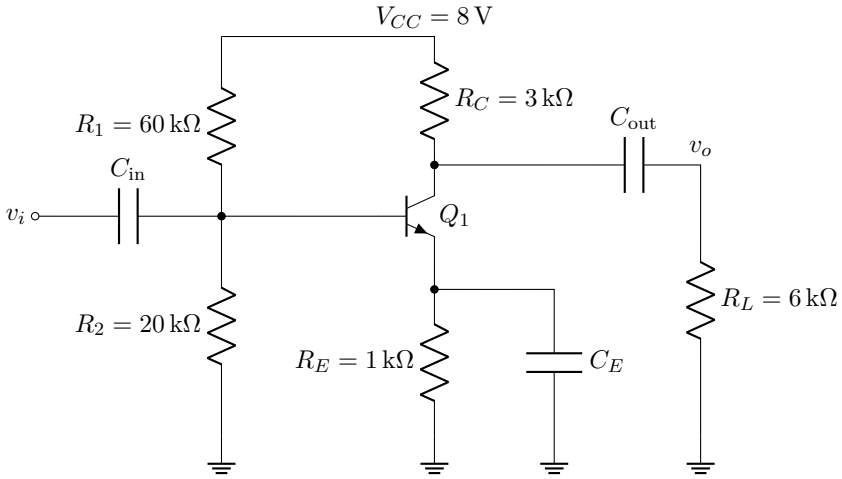


Figure 4. Divider-biased common-emitter amplifier.

(a) Explain why an NPN BJT with a forward-biased base–emitter junction and reverse-biased base–collector junction can carry collector current. Identify the dominant transport mechanism through the neutral base. **[1]**

(b) Replace the divider by its Thévenin equivalent. Calculate I_B, I_C, V_B, V_E, V_C and V_{CE}, retaining base-current loading. Verify forward-active operation using the base and collector voltages. **[3]**

(c) With C_E connected, draw the midband hybrid-π equivalent circuit. Calculate g_m, r_π, $A_v = v_o/v_i$, R_{in} seen to the right of C_{in}, and R_{out} with R_L removed and $v_i = 0$. **[2]**

(d) Remove C_E. Derive the new gain and input resistance, retaining finite β. State whether the DC operating point changes. **[2]**

Parts (e)–(f) are independent of Figure 4.

(e) An emitter follower has $g_m = 40$ mS, $\beta = 100$, $r_o = \infty$, and $R_E = 1\,\text{k}\Omega$. Its collector is at AC ground; a coupling capacitor connects a $1\,\text{k}\Omega$ load to its emitter. An ideal bias current source feeds its base, and a signal source of resistance $R_s = 3\,\text{k}\Omega$ drives it. Find its *loaded* base-to-emitter gain and input resistance, and its output resistance with the load removed and source voltage set to zero. **[2]**

(f) For a common-base stage with $I_C = 1$ mA, $\beta = 100$, $r_o = \infty$, an ideally AC-grounded base, and effective collector load $2\,\text{k}\Omega$, find the gain from emitter voltage to collector voltage and the resistance looking into the emitter (no external emitter resistor). **[1]**

End of paper — Total: 30 marks